PERIODIC WILSON STATES AND REFLECTION POSITIVITY

Abstract. We study finite-period Gibbs states of the four-dimensional Wilson SU(2) model at fixed lattice spacing. A transfer-kernel comparison and an exact conditional integration give a common period for these states. Reflected local perturbations then compare their expectations with those of reflection-positive states. Compact convex separation yields reflection positivity for the entire periodic class, including the coordinate and diagonal lattice mirrors and bounded Borel half-space observables. We also derive domination by finite chessboard seminorms.

Contents

1. Introduction 1
2. Wilson measures and local limits 2
3. Finite reflection forms 4
4. Chessboards and a pressure comparison 5
5. Bond reflections and a common period 7
6. Reflection positivity of periodic Gibbs states 11
7. Domination by finite chessboard seminorms 13
References 14

1. Introduction

Finite Wilson measures admit reflection inequalities on suitable periodic lattices. Passing such an inequality to a subsequential infinite-volume limit preserves it for that limit. A periodic Gibbs state is instead defined by its local conditional distributions and a finite-index group of translation periods. These conditions do not identify it with a chosen limit of finite periodic measures. The question is whether the reflection inequalities hold throughout this periodic class.

The proofs below separate the symmetry of the states from the positivity of their reflection forms. A contraction of the transfer kernel first controls an antisymmetric observable on two adjacent layers. Exact integration against the Gibbs conditional density extends this comparison to observables on arbitrarily many layers. The resulting bond-reflection symmetry supplies a period common to all periodic states. With this period fixed, folding a repeated local source produces a reflection-positive comparison state. A separating local observable would contradict this comparison if a periodic state failed the reflection inequality.

The finite-volume action has the compact-link form used in Bałaban's lattice construction [Ba87, pp. 251–252]. The state comparison is proved here from the conditional kernels, the positive convolution spectrum and finite reflection forms. No assertion of uniqueness of the Gibbs state is needed. The arguments hold at fixed spacing and fixed coupling. Their boundary estimates may depend on the spacing. They provide no continuum limit or mass-gap conclusion.

Constants denoted by C may change between estimates. Their dependence is restricted to the fixed coupling, the lattice geometry and the specified finite supports unless additional parameters are displayed. All Haar measures have mass one.

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2. Wilson measures and local limits

Let $a > 0$ be the lattice spacing, let $\beta \geq 0$ be the coupling, and let $G = \mathrm{SU}(2)$. We use lattice units in the proofs, so the lattice is $\mathbb{Z}^4$ and the physical position of x is ax. A link field assigns $U_\mu(x) \in G$ to the positive link $(x, x + e_\mu)$. The reversed link has value $U_\mu(x)^{-1}$. On a torus we retain the base and direction as the link index. For $\mu < \nu$ the plaquette $p = (x; \mu, \nu)$ is oriented first along e_μ and then along e_ν. Let U_p be its ordered boundary product and set

$$a_p(U) = 1 - \tfrac{1}{2} \operatorname{tr} U_p, \qquad w_\beta(g) = \exp\{-\beta(1 - \tfrac{1}{2} \operatorname{tr} g)\}.$$

The trace is real on G, and $0 \leq a_p \leq 2$. If Γ is a full-rank period lattice, the Wilson measure on $\mathbb{Z}^4/\Gamma$ is

$$d\mu_\Gamma(U) = Z_\Gamma(0)^{-1} \prod_p w_\beta(U_p) \prod_b dU_b. \tag{2.i}$$

The two products are indexed by plaquettes and positive links, respectively, and dU_b is normalized Haar measure. The density is strictly positive. The bound $e^{-2\beta\#\{p\}} \leq Z_\Gamma(0) \leq 1$ follows by integrating the pointwise bounds on the weight against product Haar. Multiplying numerator and denominator by the same factor $e^{\beta\,\#\{p\}}$ gives the weight $\exp(\frac{\beta}{2} \sum_p \operatorname{Re} \operatorname{tr} U_p)$. This is the compact-link Wilson action used by Bałaban [Ba87, pp. 251–252, (0.2)].

Write $\tau_v x = x + v$ for translation by $v \in \mathbb{Z}^4$. For a lattice isometry s, write $sF(U) = F(s^*U)$, where $(s^*U)(x, y) = U(sx, sy)$ on the infinite lattice. On a quotient, apply s to an oriented lattice lift of the link and then project its base and direction, inverting the variable when the direction reverses. This uses the indexed link even when two quotient links have the same endpoints. Every isometry that descends to the torus preserves μ_Γ. It permutes the Haar variables, inverting those whose orientation changes. It preserves the plaquette trace because a change of base conjugates the holonomy and reversal inverts it. Site gauges act by $U^g(x, y) = g(x)U(x, y)g(y)^{-1}$. The plaquette holonomy is conjugated at its base and Haar measure is invariant under both endpoint multiplications.

On the infinite lattice, let $\Omega = G^{\mathcal{E}_+}$ for the countable set $\mathcal{E}_+$ of positive links. A local function depends on finitely many link variables. The product topology makes Ω compact and metrizable. Continuous local functions are uniformly dense in $C(\Omega)$. To see this, enumerate the links and freeze coordinates beyond the first N. This changes the metric $\sum_j 2^{-j} \min(1, \|U_j - V_j\|)$ by at most 2^{-N}.

For a finite set D of positive link indices, let $\mathcal{P}(D)$ be the plaquettes whose boundaries use a link of D in either orientation. Let ∂D be the positive link indices used by those boundaries and lying outside D. For an exterior configuration η, the Wilson conditional density on G^D is

$$h_D(V \mid \eta) = \frac{\exp[-\beta \sum_{p \in \mathcal{P}(D)} a_p(V\eta)]}{\int_{G^D} \exp[-\beta \sum_{p \in \mathcal{P}(D)} a_p(W\eta)]\, dW}. \tag{2.ii}$$

Here $V\eta$ uses V on D and η outside it. Denote the resulting replacement kernel by γ_D. Its denominator lies between $e^{-2\beta|\mathcal{P}(D)|}$ and 1. The density is continuous in $(V, \eta|_{\partial D})$. For every continuous local F, $\gamma_D F$ is continuous and local, by uniform continuity on the finite compact coordinate space. Uniform approximation then gives $\gamma_D C(\Omega) \subset C(\Omega)$.

A Wilson DLR state is a probability ν with $\nu\gamma_D = \nu$ for all finite D. This is equivalent to $\gamma_D F$ being the conditional expectation of F given the exterior links. Indeed, for bounded exterior-measurable J one has $\gamma_D(JF) = J\gamma_D F$, and the measure equation applies to JF. The converse follows by integrating the conditional identity. A periodic state is invariant under a finite-index translation subgroup of $\mathbb{Z}^4$.

Every Wilson DLR state is invariant under site gauges. For a gauge with finite support, take D to contain all incident links. The exterior is fixed, and the Haar changes of variables on D preserve every plaquette trace in eq. 2.ii. The DLR equation therefore gives invariance under that gauge. For a general site gauge and a local function, its action depends only on gauges at the endpoints

of the finitely many links in its support. A finite-support gauge with these values gives the same integral. Continuous local functions determine the measures, so invariance extends to bounded Borel tests.

Lemma 2.1 (Local Gibbs limits)**.** Fix $a > 0$ and $\beta \geq 0$. Let Γ_i be torus period lattices whose shortest nonzero periods tend to infinity. The periodically lifted Wilson laws have weak subsequential limits on Ω. Each limit is a Wilson DLR state and retains the lattice isometries that preserve the torus laws eventually. In particular, a cofinal sequence of cubic tori yields a translation- and signed-permutation- invariant Wilson DLR state. For a fixed finite-index period subgroup Q, the Q-invariant Wilson DLR class is compact and convex.

The local DLR passage also applies after adding a fixed real periodic interaction of bounded Borel local terms, with a uniformly finite range and finitely many term types per period cell. Use tori compatible with that interaction. Along a convergent subsequence, expectations converge for every bounded Borel local test. If the interaction is multiplied by t, any weak limit of its DLR states as $t \to 0$ is Wilson DLR, with the same local Borel convergence. ■

Proof. Probability measures on a compact metric space are sequentially weakly compact. Fix D and continuous local F. The window, its interaction neighborhood and the support of F embed injectively on all sufficiently large tori. Every torus interaction meeting the window then has a unique lifted occurrence meeting D, counting the given term indices. Integrating the torus variables of D first therefore gives exactly $\widetilde{\mu}_{\Gamma_i}(\gamma_D F) = \widetilde{\mu}_{\Gamma_i}(F)$, where the tilde denotes periodic lift. The identity is local in F. The lifted law generally has other copies of the D-links in its infinite exterior. The two integrands are continuous, so the identity passes to a weak limit. Uniform approximation and the contraction of γ_D extend it to $C(\Omega)$. Thus the Borel measures $\nu\gamma_D$ and ν agree. An isometry acts continuously on Ω, so its invariance passes to the limit. The same closed affine conditions establish compactness and convexity for a fixed Q.

For a Borel perturbation, use the same local conditional identity. The conditional density on any fixed window D is bounded by the exponential of the oscillation of the sum of interaction terms meeting D. This gives a uniform bound for the marginal density relative to Haar on G^D. The bound passes to weak limits by testing against nonnegative continuous functions. For a bounded Borel function on G^D, approximation in Haar L^1 by continuous functions now gives convergence of the expectations. The perturbed kernel preserves bounded Borel locality, so its local conditional identities pass to the limit. Equality of the resulting probability measures gives the full DLR equations.

If the unscaled perturbation terms meeting D have total supremum bound C_D, the ratio of the perturbed to Wilson conditional density lies between $e^{-2|t|C_D}$ and $e^{2|t|C_D}$, uniformly in the exterior. The conditional identities thus pass to the Wilson kernel as $t \to 0$, and the marginal bound is uniform for bounded t. □

For a fixed finite bond window D, the marginal density on D of an unperturbed Wilson law on a sufficiently large torus and its lift is $\int h_D(V \mid \eta)\, d\widetilde{\mu}_\Gamma(\eta)$, by the local conditional identity. The same formula holds in a Wilson limit. As V ranges over G^D, the continuous functions $h_D(V \mid \cdot)$ form a compact family in supremum norm. Weak convergence and a finite uniform net give uniform convergence of these marginal densities along a convergent subsequence. They all have the lower bound $e^{-2\beta|\mathcal{P}(D)|}$, so the ratios converge uniformly to one. Consequently, along a fixed cofinal family, the infimum of the uniform relative marginal error over its torus limit points tends to zero. Otherwise a violating subsequence would have a weak limit in the same class and satisfy the preceding uniform convergence. These statements concern fixed a, β, D.

3. Finite reflection forms

The reflection with normal n at height c is

$$\theta_c x = x - \frac{2(n \cdot x - c)}{|n|^2} n, \qquad \Theta_c F = \overline{\theta_c F}.$$

We take $n = e_\mu$ or $e_\mu \pm e_\nu$ with $\mu < \nu$. Preservation of $\mathbb{Z}^4$ requires and is implied by $c \in \frac{1}{2}\mathbb{Z}$ in the coordinate case and $c \in \mathbb{Z}$ in the diagonal case, by the displayed formula. There are four coordinate and twelve diagonal orientations. A coordinate cut is a site cut at integral c and a bond cut at half-integral nonintegral c. In physical units the height offsets are multiplied by a. On the infinite lattice a closed-half algebra consists of bounded Borel functions of links with both endpoints in that half.

For $q = |n|^2$ a split period lattice has the form

$$\Gamma = \Gamma_\parallel \oplus \mathbb{Z}(2zn/q), \qquad \Gamma_\parallel \subset \mathbb{Z}^4 \cap n^\perp, \qquad \operatorname{rank} \Gamma_\parallel = 3. \tag{3.i}$$

Here z is a positive integer and the normal period vector is integral. The two cuts on the torus have heights $c, c + z$ modulo $2z$. The reflection interchanges the two intervening strips and fixes their common sites. At height $c + z$ the displacement is exactly one normal period. On a torus a half-space link must have a lift lying wholly in the corresponding closed strip. This geometric convention distinguishes the two links of a shortest periodic circle even when they have the same projected endpoints. For diagonal cuts we take $z \geq 3$ and require every nonzero tangential period to have Euclidean length greater than 2. A plaquette has diameter $\sqrt{2}$, so its vertices remain distinct under this quotient. The height ranges of the plaquettes crossing the two cuts are disjoint modulo the normal height period. Coordinate cuts allow all $z \geq 1$.

Proposition 3.1 (Finite reflection positivity)**.** Let $\beta \geq 0$. On a split torus as above the Wilson law is reflection invariant and

$$\mu_\Gamma(\Theta_c F \, F) \geq 0$$

for every bounded complex Borel function on either closed half. This includes both coordinate cut types on even coordinate tori. If E is bounded, real and measurable on one closed half, the normalized law with additional density $e^{E + \theta_c E}$ has the same two properties. ∎

Proof. Write $s(g) \in S^3 \subset \mathbb{R}^4$ for the unit quaternion coordinates of $g \in G$. Then $\frac{1}{2}\operatorname{tr}(gh^{-1}) = s(g) \cdot s(h)$, and

$$w_\beta(gh^{-1}) = e^{-\beta} \sum_{m \geq 0} \frac{\beta^m}{m!} \sum_{i_1, \ldots, i_m = 1}^{4} \prod_{j=1}^{m} s_{i_j}(g) \prod_{j=1}^{m} s_{i_j}(h). \tag{3.ii}$$

This is a positive Gram expansion in real bounded features. The coefficients sum to $e^{3\beta}$, so the expansion and its finite products converge absolutely and uniformly.

At a coordinate site cut no plaquette meets both strict halves. The plaquettes in the two halves occur in reflected pairs, and the plaquettes lying on the cuts depend only on the shared links. When the normal period is two, every plaquette with normal edges still lies in one strip by its geometric lift, even though its vertices lie on the cuts. Conditional on the shared links, the reflection form is an absolute square integrated against a positive measure.

At a diagonal cut use a signed permutation to take $n = e_1 + e_2$. A link changes height by 0 or 1 in absolute value, so no link joins strict opposite sides. The only crossing plaquettes lie in the $(1, 2)$ plane and have vertex heights $c - 1, c, c + 1, c$ in cyclic order. Orient the two paths P_+, P_- joining the two height-c vertices in the same direction. The boundary holonomy is $P_+ P_-^{-1}$ up to base change and reversal. The reflection exchanges the two paths, and eq. 3.ii applies. The links within the cuts are fixed pointwise and can be conditioned on. All remaining plaquettes lie in one closed half or in the cuts. Apply this argument at both cuts.

At a coordinate bond cut let X, Y be the parallel link arrays in the two adjoining layers, identified by reflection, and let T_x be the crossing link from the X layer to the Y layer at x. Reverse a crossing link when necessary to give it this orientation, which preserves its Haar measure. Each crossing plaquette has trace equal to that of $X_{xy}T_yY_{xy}^{-1}T_x^{-1}$. For $(T \cdot Y)_{xy} = T_xY_{xy}T_y^{-1}$, its kernel before integrating the crossing links is $C(X, T \cdot Y)$, where

$$C(X, Y) = \prod_{xy} w_\beta(X_{xy}Y_{xy}^{-1}).$$

The function C is invariant under simultaneous layer gauges, since $(S\cdot X)_{xy}(S\cdot Y)_{xy}^{-1}$ is the conjugate of $X_{xy}Y_{xy}^{-1}$ by S_x. Haar invariance therefore gives

$$\int C(X, T \cdot Y)\, dT = \iint C(S \cdot X, T \cdot Y)\, dS\, dT.$$

The product Gram expansion of C now averages each feature in its own argument. Thus the integrated crossing kernel is again a positive Gram sum. The crossing links are independent Haar variables before the crossing factors are integrated. They enter neither half-space tests nor the one-side factors. The crossing variables at the two cuts are distinct, so their kernels multiply. For a coordinate period of two, the two layers may occur at both cuts, but the crossing links at those cuts still have distinct indices and the product Gram argument still applies.

In each case, after identifying the two halves by reflection, the unnormalized pairing of F, H is a sum of terms $b_j \int \overline{I_j(F; Y_0)} I_j(H; Y_0)\, d\lambda(Y_0)$ with $b_j \geq 0$. Here Y_0 is the shared link array, λ includes its plaquette factors, and I_j integrates one half against its plaquette factors and the corresponding Gram feature. There is no shared link array at bond cuts. The coefficient bound and boundedness of the tests justify Fubini. This proves positivity and Cauchy–Schwarz for the reflection pairing. Reflection invariance follows from the lattice isometry in eq. 2.i. For the perturbed law, insert Fe^E into the pairing. Its additional density is reflection invariant. □

For a cofinal sequence of even cubic tori, every fixed continuous local test in an infinite coordinate half-space eventually lies in the corresponding torus half with the same link indices. Its reflection inequality passes to any weak limit. More generally this passage holds for each fixed mirror on cofinal split tori, and for the reflected perturbations of Proposition 3.1. The following lemma completes the half-space algebra for these limits and for the periodic states considered later.

Lemma 3.2 (Completion of the half-space algebra)**.** Let ν be a probability measure on Ω invariant under an admissible reflection θ_c. If $\nu(\Theta_c FF) \geq 0$ for every continuous local function of the links in the closed positive half-space, the same inequality holds for every bounded Borel function of those links and for every bounded Borel function of the closed negative half-space. ■

Proof. Continuous functions are dense in L^2 of the positive-half marginal by regularity on its compact metric configuration space. Continuous cylinder functions approximate them uniformly. Thus each bounded Borel half-space function F has continuous local approximants $F_j \to F$ in $L^2(\nu)$. Reflection invariance makes Θ_c an antilinear isometry. Cauchy–Schwarz gives

$$|\nu(\Theta_c F_jF_j) - \nu(\Theta_c FF)| \leq \|F_j - F\|_2(\|F_j\|_2 + \|F\|_2),$$

so positivity passes to F. A test on the negative half is the reflected conjugate of a positive-half test. Its reflection form has the same value, which proves the other assertion. □

4. Chessboards and a pressure comparison

Lemma 4.1 (One-direction chessboard inequality)**.** In a split torus let the normal height period be $2MH$, with positive integers M and H. Cut it into $2M$ slabs with faces $c + kH$, each of an admissible type and with the finite reflection forms of Proposition 3.1. Suppose f_k is a bounded complex Borel function of links whose lattice lifts lie in closed slab k. Copy a slab function using

the unique element of the group generated by the face reflections that carries its slab to each destination. Conjugate the function when that element reverses orientation. Write $d(f_k)$ for the Wilson expectation of this disseminated product. Then $d(f_k) \geq 0$ and

$$\left| \mu_\Gamma \left(\prod_{k=0}^{2M-1} f_k \right) \right| \leq \prod_{k=0}^{2M-1} d(f_k)^{1/(2M)}. \tag{4.i}$$

■

Proof. Reflection in face j sends the slab index k to $2j-1-k$ modulo $2M$. The generated group acts freely and transitively on slabs. Its translations change indices by even integers, and its reflections send an even index to an odd one. The corresponding maps of the torus satisfy these relations, since $2MH$ is its normal height period. This gives unique copies from a reference slab, with conjugation on the reflected copies. A constant dissemination is the product of a function on one half with its conjugated reflection, so it has nonnegative expectation. Shared links may occur in two slab functions. Both halves permit those links and the same identity applies.

Transport the $2M$ profiles to the reference slab. For a filling σ of the slabs by these profile labels, divide its expectation by the product over slabs of $(d(f_{\sigma(k)}) + \epsilon)^{1/(2M)}$, where $\epsilon > 0$. Call the resulting number $T(\sigma)$. Reflection Cauchy–Schwarz gives

$$|T(\sigma)|^2 \leq T(\sigma_+)T(\sigma_-),$$

where $\sigma_\pm$ retain one semicircle each and reflect it to fill the other. The denominator factors for the two reflected fillings together repeat each original factor twice. Both reflected values are nonnegative. Choose a filling of maximal absolute value. If the maximum is positive, both reflections of this filling again attain it. Start at the first face of any constant cyclic run and retain the half beginning there. If the run has length less than M, the retained half and its reflection have a run of twice that length. Once the run has length at least M, retain a full semicircle of it to get a constant maximizing filling. A constant normalized filling is at most one. Thus the maximum is at most one, also when it is zero. Let $\epsilon \downarrow 0$. □

Corollary 4.2 (Cubic chessboards)**.** Let $\Gamma = \bigoplus_{\mu=1}^{4} N_\mu \mathbb{Z} e_\mu$, where the positive integers N_μ are the coordinate periods. Tile $\mathbb{Z}^4/\Gamma$ by congruent axis-aligned cells whose width in direction μ divides N_μ, giving an even number n_μ of cells in that direction. Place their faces at coordinate bond cuts. A cell function depends only on links wholly within that cell. Let $N = \prod_\mu n_\mu$, and define the chessboard seminorm of a cell function by

$$p_\Gamma(F) = \mu_\Gamma \left(\prod_C F^{[C]} \right)^{1/N},$$

where the copies are disseminated in the four coordinate directions. For bounded complex Borel cell functions F_C,

$$\left| \mu_\Gamma \left(\prod_C F_C \right) \right| \leq \prod_C p_\Gamma(F_C). \tag{4.ii}$$

Moreover $p_\Gamma(1) = 1$, $p_\Gamma(\bar{F}) = p_\Gamma(F)$, and p_Γ is absolutely homogeneous and subadditive. ■□

Proof. Apply Lemma 4.1 first to the products in the slabs of direction one. Each resulting expectation is the dissemination of one such slab. Regroup its factors by their slabs in direction two and apply the same lemma. Reflections in distinct directions commute, and conjugating a product conjugates its factors. Thus the second step copies the previous pattern in the second direction, with the product of the two root exponents. Repeat in directions three and four. Each original cell then has exponent $1/N$. The fully disseminated expectation is nonnegative by splitting at any pair of opposite faces. Exactly half the copies are conjugated, giving absolute homogeneity. Conjugating the entire nonnegative expectation gives the conjugation identity. Expand the dissemination

of $F+G$. Applying eq. 4.ii to each term gives the binomial bound $(p_\Gamma(F)+p_\Gamma(G))^N$. Taking the root proves subadditivity. □

Lemma 4.3 (Gibbs lower tangent)**.** Let $Q \subset \mathbb{Z}^4$ have finite index and let A be a bounded real Borel local observable. Let $\Gamma_i \subset Q$ have rectangular fundamental parallelepipeds D_i with all side lengths tending to infinity. Put $\kappa = [\mathbb{Z}^4 : Q]^{-1}$ and $V_i = [\mathbb{Z}^4 : \Gamma_i]$. Let $Z_i(t)$ be the integral of the unnormalized Wilson weight multiplied by $\exp(t\sum_{v\in Q/\Gamma_i}(\tau_v A)^{\Gamma_i})$, with the superscript denoting periodization of the observable. For every Q-invariant Wilson DLR state ν and fixed real t,

$$V_i^{-1}\log\frac{Z_i(t)}{Z_i(0)} \geq t\kappa\,\nu(A) - o(1). \tag{4.iii}$$

The error is uniform in ν. The statement applies to the rectangular split tori in eq. 3.i, including diagonal normals. ■

Proof. Take D_i half-open, so its lattice vertices represent the torus classes once. Let $\mathcal{L}_i$ consist of the positive links based at those vertices. This identifies its Haar variables with the torus link variables. Since $\Gamma_i \subset Q$, the points of $Q \cap D_i$ represent Q/Γ_i. Let W_i sum over those finitely many anchors whose translated source supports are strictly interior to the window. This convention also includes constant sources, whose supports are empty. With the exterior links η fixed, Jensen's inequality for the conditional Wilson law gives

$$\log\gamma_{\mathcal{L}_i}(e^{tW_i})(\eta) \geq t\gamma_{\mathcal{L}_i}W_i(\eta),$$

The logarithm on the left is an open-boundary partition ratio. Away from a collar of fixed thickness the torus and open plaquette and source terms agree as functions of the common Haar variables. In the collar their absolute differences have sum at most $C(\beta+|t|\|A\|_\infty)|\partial_R D_i|$, where $|\partial_R D_i|$ counts the sites in a collar of thickness R on both sides of the boundary. Each plaquette term is bounded by 2β and each fixed translated support has bounded incidence per link. Comparing the two numerators and denominators by these pointwise bounds gives the same order of error for the logarithmic ratios, uniformly in η.

There are κV_i torus source copies, and all but $O(|\partial_R D_i|)$ occur in W_i. Integrating the Jensen bound against ν, the DLR equation and Q-invariance give

$$\nu(\gamma_{\mathcal{L}_i}W_i) = \nu(W_i) = (\kappa V_i + O(|\partial_R D_i|))\nu(A).$$

Finally V_i is the Euclidean volume of D_i. Disjoint unit cubes centered at the collar lattice sites lie in a collar enlarged by a fixed constant, whose volume divided by V_i tends to zero as every side length tends to infinity. This also proves the claim for tilted rectangular boxes. □

5. Bond reflections and a common period

The period of a periodic Gibbs state may initially depend on the state. The next theorem replaces it by a common period. The proof uses a perturbation of the finite transfer kernel to compare adjacent layers, then uses the conditional law to remove outer layers of an arbitrary finite observable.

Theorem A (Bond-reflection symmetry)**.** At spacing $a>0$ and coupling $\beta>0$, every periodic Wilson DLR state is invariant under reflection in every coordinate bond mirror $x_\mu = a(i+\frac{1}{2})$, where $i\in\mathbb{Z}$. In particular it is invariant under all translations in $2a\mathbb{Z}^4$. ■□

Assume $\beta>0$ throughout this section and fix one coordinate as time. Section 2 proves gauge invariance of every Wilson DLR state. A reflected state has the same property, since reflection preserves the Wilson specification.

5.1. **The transfer kernel.** Let C be the convolution operator on $L^2(G)$ with kernel $w_\beta(uv^{-1})$. This kernel is real, symmetric, and at least $e^{-2\beta}$. For each integer $d \geq 1$, let Π_d be the Peter–Weyl projection for the irreducible representation of dimension d. Convolution by a central function acts by a scalar on each such space. The Haar class density $2\sin^2 t/\pi$ and the character $\sin(dt)/\sin t$ give

$$C = \sum_{d\geq 1} \lambda_d \Pi_d, \qquad d\lambda_d = e^{-\beta}(I_{d-1}(\beta) - I_{d+1}(\beta)), \qquad \lambda_d = \frac{2e^{-\beta}}{\beta} I_d(\beta) > 0. \tag{5.i}$$

Here

$$I_n(\beta) = \frac{1}{\pi}\int_0^\pi e^{\beta\cos t}\cos(nt)\,dt = \sum_{k\geq 0} \frac{(\beta/2)^{n+2k}}{k!(n+k)!}.$$

The series follows by expanding $e^{(\beta/2)(e^{it}+e^{-it})}$. Integration by parts in $\int_0^\pi e^{\beta\cos t}\sin t\sin(dt)\,dt$ gives the last equality in eq. 5.i. The series bounds I_{n+1}/I_n by $\beta/(2(n+1))$, while

$$\frac{I_n}{I_{n+1}} = \frac{2(n+1)}{\beta} + \frac{I_{n+2}}{I_{n+1}} \leq C_\beta(1+n).$$

Thus the ratios between indices of bounded distance grow at most polynomially in the indices. The factorial decay of λ_d, with the polynomial dimensions and derivatives of the matrix coefficients, gives trace class operators C and $C^{1/2}$ with smooth real symmetric kernels.

For a finite set $\mathcal{B}$ of positive spatial bonds use $C_\mathcal{B} = \bigotimes_{e\in\mathcal{B}} C$. Its kernel is

$$C_\mathcal{B}(u,v) = \prod_{e\in\mathcal{B}} w_\beta(u_e v_e^{-1}).$$

Simultaneous vertex gauge transformations preserve this kernel. Write $\Pi_{\boldsymbol{d}}$ for the tensor projections and $\lambda_{\boldsymbol{d}} = \prod_e \lambda_{d_e}$. For a polynomial f in matrix entries and their conjugates, define on smooth functions

$$\widehat{f}F = \sum_{\boldsymbol{d},\boldsymbol{d}'} \frac{\lambda_{\boldsymbol{d}}}{\lambda_{\boldsymbol{d}'}} \Pi_{\boldsymbol{d}'} M_f \Pi_{\boldsymbol{d}} F. \tag{5.ii}$$

The tensor-product rule for SU(2) confines each coordinate difference $|d_e - d'_e|$ to a bound determined by the degree of f in that bond. Thus each index has a bounded number of partners, and each displayed ratio grows at most polynomially. The coefficients of a smooth function decay faster than any power of the joint index, by applying powers of the Casimir in each group coordinate. The sup norm of a matrix-coefficient block and of each fixed derivative is bounded by its L^2 norm times a polynomial in the indices. This follows from orthogonality, the block dimension, and differentiation in the finite-dimensional representations. The series therefore converges in every smooth norm and defines a continuous map on smooth functions. It also acts on smooth families in finitely many additional variables, because derivatives in those variables commute with each term and satisfy the same bounds.

Multiplying the series by $C_\mathcal{B}$ gives $C_\mathcal{B}\widehat{f} = M_f C_\mathcal{B}$ on this domain. By symmetry of the kernel this is the pointwise identity

$$f(w)\int C_\mathcal{B}(v,w)F(v)\,dv = \int C_\mathcal{B}(v,w)(\widehat{f}F)(v)\,dv. \tag{5.iii}$$

The spectral definition in eq. 5.ii supplies the domain of this operation. In particular the proof uses no application of an unbounded convolution inverse to a general smooth function.

Consider a cubic torus of integer side $s \geq 2$ in lattice units, and let $\mathcal{E}$ be the spatial bonds in one layer. Let P_g average over the vertex gauge group of that layer, and let M multiply by its spatial Wilson factor. Define the transfer operator by

$$T = P_\mathrm{g} C_\mathcal{E}^{1/2} M C_\mathcal{E}^{1/2} P_\mathrm{g}.$$

It is positive and trace class. The gauge action preserves the spatial Wilson factor and commutes with the convolution, so both M and $C_{\mathcal{E}}^{1/2}$ commute with P_{g}. Across a time step the two spatial arrays u, v and the time links g give the factor $C_{\mathcal{E}}(u, g \cdot v)$. On the bond (x, y) the corresponding time plaquette has product $u_{xy} g_y v_{xy}^{-1} g_x^{-1} = u_{xy}((g \cdot v)_{xy})^{-1}$. Integrating g inserts the projection P_{g}, with no extra factor because Haar is normalized. The remaining cyclic kernel integral equals $\mathrm{Tr}(T^s)$.

Let Ξ be a bounded operator commuting with complex conjugation on $L^2(G^{\mathcal{B}})$ of norm at most one, commuting with the vertex gauge action. Suppose $C_{\mathcal{B}}^{1/2} \Xi C_{\mathcal{B}}^{1/2}$ has a continuous strictly positive kernel $k_\Xi(u, v)$. Define the real local perturbation

$$\phi_\Xi(u, v) = \log \frac{k_\Xi(u, v)}{C_{\mathcal{B}}(u, v)}, \tag{5.iv}$$

where the two spatial arrays in the original field are compared after transporting the upper one by the intervening time links. The simultaneous gauge covariance makes this a gauge-invariant observable of the original bonds.

Take a periodic DLR state ν and choose $m\mathbb{Z}^4$ in its period group, with m large enough that translates of the spatial block at a given time step are disjoint. Repeat ϕ_Ξ at those anchors on tori with s a sufficiently large multiple of m. At step j, the modified crossing operator before time-link integration is $C_{\mathcal{E}}^{1/2} \Xi_j C_{\mathcal{E}}^{1/2}$, where Ξ_j is the tensor product of the inserted contractions and identities on the other bonds. Each Ξ_j has norm at most one and commutes with P_{g}. The modified partition function is the trace of the cyclic product of $M C_{\mathcal{E}}^{1/2} \Xi_j C_{\mathcal{E}}^{1/2} P_{\mathrm{g}}$. The trace formulas follow by splitting the cyclic integral into two Hilbert–Schmidt kernels. Cyclicity of the trace and commutation with P_{g} rewrite the product as $\Xi_1 T \cdots \Xi_s T$ inside the trace. The Schatten Hölder inequality, with $\|\cdot\|_{\mathfrak{S}_s}$ the Schatten s-norm, gives

$$0 < Z_s(\phi_\Xi) = \mathrm{Tr}(\Xi_1 T \cdots \Xi_s T) \le \prod_{j=1}^{s} \|\Xi_j T\|_{\mathfrak{S}_s} \le \|T\|_{\mathfrak{S}_s}^s = Z_s(0). \tag{5.v}$$

The positivity at the left comes from the kernels in the modified integral, and the final equality uses positivity of T. Lemma 4.3 with period $m\mathbb{Z}^4$ now gives $\nu(\phi_\Xi) \le 0$.

5.2. **The comparison of two adjacent layers.** Let $Y(u, v)$ be real and continuous on the two arrays of a finite spatial block $\mathcal{B}$. Suppose it is invariant under a common vertex gauge and antisymmetric under exchange of the arrays. As in eq. 5.iv, transport the upper array by the intervening time links. This defines the local observable of the original bond field whose expectation is denoted by $\nu(Y)$. We claim

$$\nu(Y) = 0. \tag{5.vi}$$

First suppose $K(u, v) = Y(u, v) C_{\mathcal{B}}(u, v)$ is a polynomial. It is the kernel of a finite-rank real skew-adjoint operator. Choose a finite sum of joint Peter–Weyl spaces containing the support in both kernel variables. On this space define $D = C_{\mathcal{B}}^{-1/2} K C_{\mathcal{B}}^{-1/2}$, using the positive eigenvalues in eq. 5.i, and extend by zero on the orthogonal complement. The extension is real, skew-adjoint and commutes with the vertex gauge action. Hence $\Xi_\varepsilon = e^{\varepsilon D}$ is an isometry for either sign of the real parameter ε. Its sandwiched kernel is $C_{\mathcal{B}} + \varepsilon K + O_K(\varepsilon^2)$ uniformly, because the difference has finite spectral support. Strict positivity of $C_{\mathcal{B}}$ admits the kernel when $|\varepsilon| < \varepsilon_0(K)$. Here the remainder bound and the positive radius depend on the fixed polynomial kernel K, as well as the fixed coupling and block. The logarithm in eq. 5.iv therefore gives $\phi_{\Xi_\varepsilon} = \varepsilon Y + O_K(\varepsilon^2)$ uniformly. The pressure comparison at each fixed ε gives $\varepsilon \nu(Y) \le C_K \varepsilon^2$. Let ε tend to zero through both signs.

For general Y, approximate $Y C_{\mathcal{B}}$ uniformly by real matrix-coordinate polynomials. Average each approximant under the common vertex gauge group and take its antisymmetric part. Both operations preserve polynomiality and uniform convergence. Division by the strictly positive kernel

yields functions to which the polynomial case applies, converging uniformly to Y. This proves eq. 5.vi.

5.3. **Removing the outer layers.** On a finite stack of layers $[j_-, j_+]$, write $t_{j,x}$ for the time link from (j,x) to $(j+1,x)$, and U_j for the spatial array of layer j transported to the lowest layer. Explicitly, at vertex x use $g_{j,x} = t_{j_-,x} \cdots t_{j-1,x}$ and act on the spatial bonds by g_j at their endpoints. In these coordinates the time links of the stack are the identity. A gauge-invariant polynomial observable of the stack is therefore a polynomial of the transported spatial arrays invariant under a common vertex gauge. Only finitely many vertices are needed for any local test.

At an interior layer j, resample the original spatial bonds of a finite block $\mathcal{B}$, conditioning on all other bonds. The time links are fixed in this resampling, so the substitution to the transported variables preserves Haar. With $w = U_{j-1}|_{\mathcal{B}}$, $v = U_j|_{\mathcal{B}}$ and $w' = U_{j+1}|_{\mathcal{B}}$, the conditional density is proportional to

$$C_{\mathcal{B}}(w,v)e^{-V(v;\eta)}C_{\mathcal{B}}(v,w')\,dv. \tag{5.vii}$$

The two kernels account for the time plaquettes through these spatial bonds. The potential V is β times the sum of the spatial plaquette actions meeting the block, and η denotes its same-layer exterior variables. This list exhausts the plaquettes in the finite-bond conditional density.

For a polynomial $f(w)$ and a smooth local factor Φ depending on the transported variables from layer j upward, put

$$\Phi^\flat = \frac{\widehat{f}_v\big[e^{-V(v;\eta)}C_{\mathcal{B}}(v,w')\Phi\big]}{e^{-V(v;\eta)}C_{\mathcal{B}}(v,w')}. \tag{5.viii}$$

The denominator is positive on the compact variable space. The smooth transfer operation leaves $\Phi^\flat$ smooth and confines its dependence to finitely many variables from layer j upward. The new support can include same-layer neighbours and the block in layer $j+1$. The identity eq. 5.iii in the conditional numerator of eq. 5.vii gives $\nu(f(w)\Phi) = \nu(\Phi^\flat)$. The normalizing factor is the same on both sides of that conditional equality. The reflected construction removes the upper outer layer.

Consider a stack of $2k+2$ layers symmetric about a bond mirror, where $k \geq 1$. Express a real invariant polynomial of the stack as a sum of terms $f(w)h(z)\Phi$, using the outer arrays w, z and a common finite spatial block. Write u, v for the first and last retained arrays. Applying the lower then the upper identity to a term gives the expectation of

$$\Upsilon^{-1}\widehat{f}_u\widehat{h}_v(\Upsilon\Phi). \tag{5.ix}$$

When u, v are nonadjacent, Υ is the product of their spatial factors and their two kernels toward the interior. For adjacent u, v, the lower removal divides by $e^{-V(u;\eta)}C_{\mathcal{B}}(u,v)$. The upper conditional numerator multiplies by $e^{-V(v;\eta')}C_{\mathcal{B}}(u,v)$ before $\widehat{h}_v$ acts. The common kernel cancels in these outside factors. The remaining multipliers independent of the operator variables can be moved through the operators, which commute on the distinct arrays. Thus in this case

$$\Upsilon = e^{-V(u;\eta)}e^{-V(v;\eta')}C_{\mathcal{B}}(u,v).$$

This explains why the common kernel enters once. For a fixed block, the expression is linear on the tensor product of the two outer polynomial spaces and the inner smooth functions. These are spaces of functions on independent variables, so the expression respects any decomposition of the original polynomial into products. Reversing the order of the layers exchanges the two operator factors and preserves the corresponding Υ, including its neighbour variables. The two-sided elimination therefore commutes with reversal of the transported array.

Average the output of eq. 5.ix over a common vertex gauge on its finite set of spatial sites. Its expectation stays the same, since this action can be realized by ordinary gauge transformations at the lowest original layer. The averaged output no longer depends on the choice of lowest layer for temporal transport. Indeed, changing the base to the first retained layer applies one common gauge to all retained arrays. Under the original bond reflection the transported array is reversed

and transformed by the common gauge $g_{j_+}^{-1}$. Both statements follow by multiplying the temporal links in the corresponding order. Gauge invariance of the averaged output removes these common actions pointwise. We obtain a continuous gauge-invariant observable of the retained stack, with the same expectation as the original polynomial and with elimination commuting with the original reflection.

Proof of Theorem A. For a gauge-invariant polynomial on the two layers adjacent to the mirror, reflection exchanges the transported arrays up to a common gauge. The difference of the two values is antisymmetric and has zero expectation by eq. 5.vi. For an invariant polynomial on $2k+2$ layers, the two-sided identity and its reflected version reduce the expectation comparison to a continuous invariant function on $2k$ layers. Matrix-coordinate polynomials are uniformly dense among continuous functions on the finite product of groups by Stone–Weierstrass. Averaging the approximants over the relevant vertex gauge group gives uniform density within the invariant functions. Induction on k therefore proves reflection invariance of the expectations of invariant continuous local tests.

Every finite local support fits in a finite even stack symmetric about the fixed bond mirror. For a general continuous local test, average over the vertex gauges of its finite support. The state and its reflected state have the same expectations before and after this averaging. The averaged test is invariant, so the two states agree on continuous local tests. These determine the measure on the product space. The time direction and the bond mirror were arbitrary. Two consecutive parallel bond reflections compose to translation by two lattice steps, which gives the claimed common period in physical units. □

6. Reflection positivity of periodic Gibbs states

Work in lattice units. By Theorem A, the periodic Wilson DLR states form the class $\mathcal{G}_\beta^P$ of Wilson DLR states invariant under $P = 2\mathbb{Z}^4$. A folded source will produce states in this class that dominate the expectation of a prescribed local test.

Lemma 6.1 (Domination by a reflected state)**.** Fix $\beta > 0$ and an admissible coordinate or diagonal mirror with normal n and height c. For every bounded real Borel function A of finitely many bonds and every $\nu \in \mathcal{G}_\beta^P$, there is $\rho \in \mathcal{G}_\beta^P$ with $\rho(A) \geq \nu(A)$. The state ρ is invariant under θ_c and reflection positive on continuous local functions of bonds in the closed positive half-space. ■

Proof. Choose $b \in P$ so that $B = \tau_b A$ depends only on bonds strictly above the mirror. Such a shift exists because n has a coordinate equal to 1. Choose an even integer $h > 0$ large enough that the endpoints of those bonds lie below height $c + H$, where $w = hn$ and $H = n \cdot w = h|n|^2$. Put

$$P_\parallel = P \cap n^\perp, \qquad Q = P_\parallel \oplus \mathbb{Z}w, \qquad Q_f = P_\parallel \oplus \mathbb{Z}(2w).$$

These are subgroups of P, with Q of finite index and $[Q : Q_f] = 2$. The tangential lattice $P_\parallel$ has an orthogonal integer basis. For a coordinate normal use the other three doubled coordinate vectors. For $n = e_\mu \pm e_\lambda$ use $2(e_\mu \mp e_\lambda)$ and the two doubled coordinate vectors outside that pair.

Take the torus with period lattice $\Gamma_{M,J} = JP_\parallel \oplus \mathbb{Z}(2Mw)$, for sufficiently large positive integers M, J, and write $\Gamma = \Gamma_{M,J}$. Choose this cofinal tail so that the explicit diagonal period bounds preceding Proposition 3.1 hold whenever needed. It has $2M$ slabs of height H between faces $c + kH$. All its side lengths tend to infinity as $M, J \to \infty$. For a local function D write D^Γ for its evaluation on the periodic lift of a torus configuration. Consider the unfolded and folded sums

$$S_{\mathrm{u}} = \sum_{v \in Q/\Gamma} (\tau_v B)^\Gamma,$$

$$S_{\mathrm{f}} = \sum_{v \in Q_f/\Gamma} \left((\tau_v B)^\Gamma + (\tau_v \theta_{c+H} B)^\Gamma \right).$$

The first places translated copies in all slabs. The second places translated copies in the even slabs and reflected copies in the odd slabs. It is invariant under Q_f and all face reflections. In fact translations by $2w$ and reflection at height $c+H$ generate the normal action used to place its terms.

Let $Z_{\mathrm{u}}(t)$ and $Z_{\mathrm{f}}(t)$ be the Wilson partition functions on this torus with added exponents tS_{u} and tS_{f}, respectively. To apply Lemma 4.1, use the slab function

$$f_k = \exp\left[t \sum_{v \in P_{\parallel}/JP_{\parallel}} (\tau_{v+kw} B)^{\Gamma}\right]$$

in slab k. Its dissemination is a normal translate of the folded weight. The unperturbed torus measure is invariant under that translation, so each disseminated partition function is $Z_{\mathrm{f}}(t)$. The chessboard inequality gives

$$Z_{\mathrm{u}}(t) \le Z_{\mathrm{f}}(t). \tag{6.i}$$

The same finite comparison holds when B depends on bonds of the closed slab, including bonds on its faces: the slab chessboard lemma permits these shared variables and retains each source occurrence with its own slab index. For a test A straddling the mirror, the unfolded partition function of its repeated array equals $Z_{\mathrm{u}}(t)$ by the global torus translation by b.

Let $\mu_{\Gamma,t}$ denote the normalized folded measure, $N_\Gamma = [\mathbb{Z}^4 : \Gamma]$, and $\kappa = [\mathbb{Z}^4 : Q]^{-1}$. Define

$$p_\Gamma(t) = N_\Gamma^{-1} \log\left(\frac{Z_{\mathrm{f}}(t)}{Z_{\mathrm{f}}(0)}\right).$$

Finite-volume differentiation gives $p'_\Gamma(t) = \kappa \mu_{\Gamma,t}(B^\Gamma)$. Indeed the κN_Γ source terms all have the same mean by Q_f periodicity and reflection at height $c+H$. Convexity and $p_\Gamma(0) = 0$ imply $p_\Gamma(t) \le t p'_\Gamma(t)$ for $t > 0$. Apply Lemma 4.3 to the unfolded array, using $\Gamma \subset Q \subset P$. For every fixed $t > 0$,

$$t\kappa\nu(B) - o(1) \le N_\Gamma^{-1} \log \frac{Z_{\mathrm{u}}(t)}{Z_{\mathrm{u}}(0)} \le p_\Gamma(t) \le t\kappa \mu_{\Gamma,t}(B^\Gamma), \tag{6.ii}$$

where the error tends to zero as $M, J \to \infty$.

At each pair of opposite slab faces the folded exponent has the form $E + \theta E$, with E its contribution from the slabs in one closed half and θ the reflection exchanging the halves. Thus Proposition 3.1 gives reflection positivity of $\mu_{\Gamma,t}$ at these faces, by inserting Fe^E in the unperturbed form. Its reflection invariance follows from the same decomposition and invariance of the unperturbed measure.

At fixed $0 < t \le 1$, apply Lemma 2.1 to the bounded finite-range folded interaction. A subsequence of $\mu_{\Gamma,t}$ converges to a perturbed DLR state ρ_t, with convergence of bounded Borel local expectations. The period Q_f and reflection invariance pass to the limit. Every continuous local test in the closed half-space of height at least c eventually lies in the corresponding torus half, so its positivity passes to ρ_t. From eq. 6.ii we have $\nu(B) \le \rho_t(B)$.

Now let $t \downarrow 0$ along a weakly convergent subsequence of these states. On a fixed finite bond set, the added exponent in the conditional density is $O(t)$ uniformly in the exterior. By Lemma 2.1, the limit ρ is a Wilson DLR state and $\rho_t(B)$ converges to $\rho(B)$ on this subsequence. The state ρ retains Q_f periodicity and the stated reflection properties, and $\nu(B) \le \rho(B)$. Theorem A upgrades its period to P. Both ρ and ν are then invariant under the original shift b, giving $\nu(A) \le \rho(A)$. The volume limit was taken at fixed t to remove the error in eq. 6.ii before division by $t\kappa$. □

Theorem B (All admissible mirrors)**.** At any fixed spacing $a > 0$ and coupling $\beta \geq 0$, every Wilson SU(2) DLR state with a finite-index period subgroup is invariant under all lattice translations and signed coordinate permutations. It is invariant and reflection positive at each mirror

$$x_\mu = c, \quad c \in \tfrac{a}{2}\mathbb{Z}, \qquad x_\mu \pm x_\lambda = c, \quad c \in a\mathbb{Z}, \quad \mu < \lambda.$$

The reflection inequality holds for every bounded Borel function of the bonds whose endpoints lie in either closed half-space. ■□

Proof. If $\beta = 0$, the DLR equation on each finite bond set forces its marginal to be product Haar. Hence the state is the unique product-Haar law. Permutation and inversion invariance of Haar give all the stated lattice symmetries. Condition on the links contained in the mirror. The remaining link sets of the two closed halves are disjoint and independent, and the reflection fixes each conditioned link. A link with exactly one endpoint on the mirror belongs to the remaining set on its side. At a bond cut the crossing links belong to neither half. Integrating them contributes one. The reflection form is therefore an integral of an absolute square. This proves positivity for continuous local tests, and Lemma 3.2 gives the Borel assertion.

Assume $\beta > 0$. In lattice units put $K = \mathcal{G}_\beta^P$. For a fixed mirror let K_c consist of the states in K invariant under θ_c and positive on the continuous local tests of the closed positive half-space. The DLR and period equations are closed affine conditions on probability measures. They can be tested by continuous cylinder functions, which the Wilson conditional kernels preserve. Hence K is compact and convex. The reflection equations and positivity inequalities on the specified tests give a closed convex subset K_c. It is nonempty by taking an unperturbed subsequential limit on the split tori.

If $\nu \in K \setminus K_c$, a weak neighborhood of ν defined by finitely many real continuous local tests is disjoint from K_c. Their joint moment vectors for K_c form a compact convex subset of Euclidean space that excludes the vector for ν. A strict separating linear functional on these vectors gives a real continuous local function A with $\nu(A) > \sup_{\rho \in K_c} \rho(A)$. This contradicts Lemma 6.1, so $K = K_c$.

Lemma 3.2 extends positivity to the bounded Borel algebras of both closed half-spaces.

The mirror was arbitrary. Composing the site and bond reflections at heights 0 and 1/2 in a coordinate direction gives a unit translation. Site reflections through the origin give the coordinate sign changes, and reflection in $x_\mu = x_\lambda$ gives the coordinate transpositions. These generate the stated symmetries. The dilation $x \mapsto ax$ restores the physical spacing and the listed mirror heights. □

7. Domination by finite chessboard seminorms

For the cutoff boards of the model, let $L \geq 3$ be odd, $a = L^{-K}$, and let the fine board side be $l = L^j \geq a$, with K a nonnegative integer and j an integer. The board faces cut lattice bonds in half. A cell function uses only bonds whose endpoints lie in the open cell. On a cubic torus of side $\ell = 2L^m$ with an integer $m \geq \max(0, j)$, the board has $N = (\ell/l)^4$ cells and an even number in every direction. Write $p_{l,\ell}$ for the seminorm of Corollary 4.2 on this board. Copies use conjugation with each orientation-reversing isometry. A cell function may be presented on any actual cell and transported to the reference cell to compute its seminorm. The result is the same if the reference cell is changed, by invariance of the torus measure and conjugation invariance of the seminorm.

Theorem C (Fixed-spacing chessboard domination)**.** Fix $\beta \geq 0$, a periodic Wilson DLR state ρ, and a finite collection of occupied cells with bounded complex Borel functions F_C on those cells. For every sequence of the compatible torus sides $\ell_k \to \infty$,

$$\left|\rho\left(\prod_C F_C\right)\right| \leq \liminf_{k\to\infty} \prod_C p_{l,\ell_k}(F_C). \tag{7.i}$$

Only the occupied cells enter the products. ■□

Proof. First let H be bounded, real and Borel on one cell. Enclose a board of side ℓ in a cube with faces on the board faces, and sum the reflected copies of H over its cells. Denote the sum by W_ℓ. Theorem B gives $\rho(W_\ell) = N\rho(H)$. Write γ for the DLR conditional kernel on the positive bonds anchored at the lattice sites of the cube. Conditional Jensen gives

$$N\rho(H) \leq \rho(\log \gamma(e^{W_\ell})).$$

Identify the anchored bonds with the torus bonds by reducing their anchors modulo the period. The conditional interaction and the torus action agree off a boundary layer. They differ in at most $C(\ell/a)^3$ plaquette terms, uniformly in the outside configuration. The source copies use only bonds in open cells, so their sum is the same function of the integrated bonds in both integrals. Comparing the numerator and denominator of the conditional exponential moment gives

$$\rho(H) \leq \log p_{l,\ell}(e^H) + C\beta \frac{l^4}{a^3\ell}. \tag{7.ii}$$

The error is divided by the number of cells. The comparison uses bounded plaquette actions and holds at fixed a, β, l.

For a bounded real cell function G, take $H = \log(1 + tG)$ with $t\|G\|_\infty < 1/2$ and $t > 0$. Then $\rho(H) \geq t\rho(G) - Ct^2\|G\|_\infty^2$. The seminorm triangle inequality and $p_{l,\ell}(1) = 1$ give

$$\log p_{l,\ell}(1 + tG) \leq t p_{l,\ell}(G).$$

Insert these in eq. 7.ii. Take the lower limit along the prescribed sequence at each fixed t and then send t to zero. This proves $\rho(G) \leq \liminf_k p_{l,\ell_k}(G)$. For a complex cell function F, choose a real phase φ so that $\rho(\mathrm{Re}(e^{-i\varphi}F)) = |\rho(F)|$. Homogeneity, the triangle inequality and $p(\overline{F}) = p(F)$ bound the seminorm of this real part by $p(F)$. We have proved eq. 7.i for a single cell.

For a finite collection of cells, choose an origin for the coarse board on fine-board faces. Choose its allowed side $l' = L^r l$, with an integer $r \geq 1$ large enough that one coarse cell contains all occupied fine cells strictly inside. The coarse origin can be shifted independently of the fine origin. The coarse faces are fine-board faces. The coarse board is compatible with every sufficiently large torus in the chosen sequence. The product $G = \prod_C F_C$ is one bounded complex function on the coarse cell. Disseminate G on the coarse board. Every copied fine factor is transported by an element of the fine-board reflection group, since reflections in coarse faces belong to that group. Its fine seminorm is therefore the seminorm of the original factor. Each occupied-cell factor occurs once in every coarse copy. The fine-board chessboard bound, followed by the root for the number of coarse cells, yields

$$p_{l',\ell}(G) \leq \prod_C p_{l,\ell}(F_C).$$

Apply the one-cell bound to G on the coarse board and take the lower limit to conclude. □

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